
A Dual-Energy Analysis of Flow Matching

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Abstract

Continuous-time flow matching models typically parameterize a vector field to transport a simple prior into an empirical data distribution. However, standard formulations rely entirely on macroscopic probability transport, forcing unconstrained neural networks to implicitly learn fundamental microscopic spatial topology from scratch. We present a rigorous mathematical framework that unifies optimal macroscopic probability transport with microscopic spatial dynamics. By framing continuous-time generative modeling as a dual-energy problem, we reconcile the global kinetic energy of optimal transport with the localized Dirichlet energy of the spatial signal. By embedding an analytical discrete spatial Laplacian into the generative vector field, we introduce Dual-Energy Flow Matching. We mathematically prove that this framework algebraically dampens high-frequency structural gradients, decoupling the required Lipschitz constant from the image resolution and establishing strictly tighter sample complexity bounds. Furthermore, we demonstrate that this deterministic formulation acts as a strict spectral normalizer, yielding a non-stiff ordinary differential equation that safely bypasses the instabilities of inverse heat models. By shifting the generative target to a physically smoothed data manifold, our framework explicitly isolates universal physical mechanics from statistical semantic intent, enabling unprecedented structural stability and representational efficiency in high-resolution pixel-space synthesis.

1 Introduction

Generative modeling has achieved unprecedented success in high-fidelity image synthesis, driven by rapid advancements in diffusion probabilistic models [24, 25, 26, 8, 5, 22, 19, 20] and continuous-time flow matching models [15, 16, 1, 7, 11]. Recently, the field has experienced a paradigm shift, transitioning from Stochastic Differential Equations (SDEs) [26] to deterministic Ordinary Differential Equations (ODEs) [15, 16, 1]. Flow matching enables straight probability paths using optimal transport principles to minimize trajectory curvature, resulting in faster sampling and more stable training dynamics. Meanwhile, neural backbones have scaled dramatically, evolving from localized U-Nets [23, 8, 5] to global diffusion transformers [19, 2, 18, 7, 11, 13, 30] capable of modeling complex high-frequency data manifolds.

The Semantic Gap. Despite these architectural milestones, a fundamental theoretical inefficiency remains, as flow matching formulates probability transport purely as a statistical curve-fitting exercise. While deep neural networks excel at learning high-level semantic abstractions, current paradigms force them to expend immense representational capacity on elementary continuous mechanics—

specifically, how signal amplitudes propagate across a 2D spatial grid to form sharp geometric boundaries. Although emerging efforts seek to inject physical priors into generative dynamics by modeling image synthesis as the reversal of thermodynamic heat dissipation [21], directly integrating the inverse heat equation is mathematically ill-posed and prone to catastrophic numerical overflow on noisy states. While recent advancements stabilize this process by embedding continuous heat trajectories into deterministic flow matching frameworks [17], flow matching treats probability flow as an unstructured point cloud, operating independently across spatial dimensions without topological bounds. This raises a critical question. If flow matching is mathematically blind to spatial topology, how do state-of-the-art models like FLUX [11] achieve coherent visual synthesis? Modern neural architectures rely universally on pre-trained Variational Autoencoders (VAEs) [10] to compress raw images into low-dimensional, low-frequency latent spaces. Under this paradigm, the VAE resolves high-frequency microscopic physics and local pixel correlations offline. Flow matching is then safely applied to these compressed latents, bypassing the steep, high-frequency gradients required to construct spatial geometry from scratch. While Just Image Transformers (JiT) [13] recently demonstrated the viability of direct pixel-space generation using simple vision transformers [6], without the VAE, it forces JiT to brute-force geometric correlations from scratch. This re-exposes a fundamental challenge. It requires an explicit mathematical mechanism to govern spatial topology when moving beyond statistical curve-fitting to generate raw pixels.

To address this, we argue that pixel-space image synthesis requires the joint governance of two distinct mathematical regimes: statistical likelihood in probability space and geometric topology in image space. By bridging the continuity equation of optimal transport with the spatial topology of Dirichlet energy dynamics, we introduce Dual-Energy Flow Matching, a framework that explicitly isolates universal physical transport from semantic hallucination. Our primary contribution is to **close this semantic gap**. We mathematically bound the metric entropy of flow matching, proving that explicitly injecting the spatial Laplacian algebraically offsets structural gradients, thereby significantly improving representational scalability and sample efficiency.

2 Related Work

Continuous-Time Generative Models. Score-based generative models [25] and denoising diffusion probabilistic models [8] established the foundation for mapping noise to data distributions through reverse-time SDEs [26]. Recent advancements simplify this paradigm by directly regressing deterministic vector fields that transport probability mass between arbitrary distributions along straight-line paths [15, 16], grounded in neural ODEs [4].

Physics-Guided Generative Priors. To reduce reliance on Gaussian noise, emerging frameworks increasingly ground generative dynamics in physical priors, such as Inverse Heat Dissipation Models (IHDMs) [21] and blurring models [9]. These blurring principles have been adapted to seamlessly bridge resolution gaps [27] and dynamically adjust non-isotropic frequency kernels to mitigate inductive biases [29]. To overcome the severe numerical instability inherent in IHDMs, Heat Dissipation Flow Matching (HDFM) integrates continuous degradation paths directly into deterministic ODE frameworks [17]. To rectify these dynamic imbalances, recent training-free frameworks introduce phase-aware energy modulation [14, 12]. By strategically boosting initial energy to overcome signal starvation and damping late-stage velocity to prevent terminal blow-up, these approaches stabilize transport and significantly enhance synthesis quality.

3 Dual-Energy Flow Matching

Notations. To formalize our analysis, we establish two distinct mathematical regimes. The *probability space* $\mathbb{R}^D$ (where $D = M^2$) is the macroscopic domain, where an entire image is treated as an unstructured state vector $\mathbf{x} \in \mathbb{R}^D$, and $p_t(\mathbf{x})$ denotes the continuous trajectory of the density. The *image space* is the microscopic domain defined explicitly on the discrete $M \times M$ spatial grid, where $\mathbf{x}_t$ is unrolled into a 2D spatial signal $\mathbf{x}_t(\mathbf{r})$ with discrete coordinates $\mathbf{r} \in \{0, \dots, M-1\}^2$. For mathematical brevity, applying an image space spatial operator to a probability space vector $\mathbf{x}_t$ implicitly denotes the physical operation applied to its 2D spatial tensor, such that $\Delta \mathbf{x}_t$ evaluates exactly to the discrete spatial Laplacian of $\mathbf{x}_t(\mathbf{r})$.

Optimal Transport Flow Matching. The goal of flow matching is to construct a probability path $p_t(\mathbf{x})$ for $t \in [0, 1]$ from Gaussian noise $p_0(\mathbf{x}) = \mathcal{N}(\mathbf{x}; 0, \mathbf{I})$ to an empirical data distribution $p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$. For a single path $\mathbf{x}_t$, it defines the continuous normalizing flow ODE $\frac{d\mathbf{x}_t}{dt} = \mathbf{v}_t(\mathbf{x}_t)$, where $\mathbf{v}_t$ is the vector field. The flow matching objective minimizes the discrepancy between the target vector field $\mathbf{v}_t$ and the estimated vector field $\mathbf{v}_\theta$ parameterized by a neural network, $\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{v}_t(\mathbf{x}_t | \mathbf{x}_0, \mathbf{x}_1)\|_2^2 \right]$. In the optimal transport flow matching framework [15, 16], the path $\mathbf{x}_t$ between a noise sample $\mathbf{x}_0$ and a data sample $\mathbf{x}_1$ is defined as a straight line $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$. Differentiating this path with respect to time t yields the optimal target vector field $\mathbf{v}_t(\mathbf{x}_t | \mathbf{x}_0, \mathbf{x}_1) = \mathbf{x}_1 - \mathbf{x}_0$. The loss function simplifies to $\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{v}_\theta(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{x}_0)\|_2^2 \right]$.

Dual Energy. The temporal evolution of the density $p_t(\mathbf{x})$ is governed by the continuity equation [28, 15, 16, 1] expressed as $\frac{\partial p_t}{\partial t} + \nabla_{\mathbf{x}} \cdot (p_t \mathbf{v}_t) = 0$, enforcing absolute mass conservation. To transport mass efficiently, optimal transport relies on the Benamou-Brenier formulation [3]. This dictates that probability mass must flow along the specific vector field that minimizes the global kinetic energy

$$E_K = \frac{1}{2} \int_0^1 \int_{\mathbb{R}^D} \|\mathbf{v}_t(\mathbf{x})\|^2 p_t(\mathbf{x}) d\mathbf{x} dt. \quad (1)$$

Minimizing this path-action integral requires a state of zero acceleration, forcing the optimal target vector field to map a straight line $\mathbf{v}^*(\mathbf{x}_t) = \mathbf{x}_1 - \mathbf{x}_0$. While this macroscopic drift is highly efficient at routing probability mass, it provides no localized driving force to geometrically correlate adjacent pixels. In continuous mechanics, the localized spatial correlation of a signal on the discrete $M \times M$ grid is quantified by its Dirichlet energy

$$E_D = \frac{1}{2} \int_0^1 \sum_{\mathbf{r} \in \{0, \dots, M-1\}^2} \|\nabla_{\mathbf{r}} \mathbf{x}_t(\mathbf{r})\|^2 dt. \quad (2)$$

The exact gradient descent flow of E_D yields the spatial Laplacian $\Delta \mathbf{x}_t$. Generative synthesis demands an active ascent of this Dirichlet energy landscape to construct sharp spatial boundaries from an initially smooth prior.

Energy Frustration and Vector Field Decomposition. A fundamental physical frustration arises because the global kinetic transport E_K minimization dictates a zero-acceleration straight line, whereas the microscopic structural force E_D dictates a curved spatial gradient flow given by $-\Delta \mathbf{x}_t$. To reconcile this, we unify the regimes into a composite Dual-Energy Flow Matching. To guarantee that this unified trajectory strictly preserves the E_K -optimal straight line, we constrain the composite vector field to the optimal transport velocity $\mathbf{x}_1 - \mathbf{x}_0$. Then, we define the velocity $\mathbf{v}_t(\mathbf{x}_t) = (\mathbf{x}_1 - \mathbf{x}_0) + \alpha_t \Delta \mathbf{x}_t$ and the corresponding ODE

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}_t(\mathbf{x}_t) - \alpha_t \Delta \mathbf{x}_t. \quad (3)$$

Therefore, the loss function becomes

$$\mathcal{L}_{\text{DEFM}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{v}_\theta(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{x}_0 + \alpha_t \Delta \mathbf{x}_t)\|_2^2 \right]. \quad (4)$$

By explicitly shifting the spatial gradients onto the analytical Laplacian, the neural network acts as an energy-compensating router, learning to pre-compensate for the physical bending of the trajectory.

4 Main Results

Spectral Sample Efficiency Bound. Flow matching assumes the prior $\mathbf{x}_0$ is Gaussian noise, forcing the neural network to synthesize the entire geometric topology from scratch. We demonstrate mathematically that decoupling the Dirichlet energy E_D from the learned semantic transport E_K allows us to rigorously bound the sample complexity and analytically derive both the schedule α_t and the prior $\mathbf{x}_0$. We establish the following theorem, with its proof provided in Appendix A.

Theorem 1. *In optimal transport flow matching on a discrete $M \times M$ spatial grid, accurately regressing the target vector field up to the spatial Nyquist limit requires the neural network's minimal spatial Lipschitz constant to scale linearly with the grid resolution $L_H = \mathcal{O}(M)$. This*

imposes an intractable sample complexity bounded by $\mathcal{O}((M/\epsilon)^{M^2})$. Under Dual-Energy Flow Matching, shifting high-frequency structural modeling onto the analytical discrete spatial Laplacian mathematically neutralizes the spectral derivative penalty at the upper frequency limits. This explicitly decouples the required Lipschitz constant L_L from the resolution M , establishing a strictly tighter and highly scalable sample complexity bound of $\mathcal{O}((L_L/\epsilon)^{M^2})$, where $L_L \ll L_H$.

ODE Spectral Normalization and Stability. With the generative ODE and its physical parameters strictly bounded, we formalize the reverse integration process. Standard flow matching models often suffer from numerical instability when generating high-frequency structures, forcing the use of tiny integration steps. We demonstrate that our analytical schedule explicitly normalizes the system, allowing for standard, highly efficient numerical solvers. We establish the following theorem, with its proof provided in Appendix B.

Theorem 2. *In optimal transport flow matching, explicitly generating high-frequency physical boundaries forces the generative ODE to become mathematically stiff. The spectral radius of its Jacobian scales as $\mathcal{O}(M^2)$, restricting explicit numerical integration step sizes to an intractable $\Delta t \leq \mathcal{O}(M^{-2})$. In Dual-Energy Flow Matching, the analytically derived schedule $\alpha_t = t/\|\omega_{\max}\|^2$ acts as a strict spectral normalizer. By bounding the physical spectral radius to $\rho \leq 1$, the composite vector field remains completely non-stiff, guaranteeing stable reconstruction with standard explicit solvers using massive step sizes $\Delta t = \mathcal{O}(1)$, entirely decoupled from the grid resolution.*

Constructing the Topological Prior. Because the mathematical guarantees of Theorem 1 strictly require $T = \|\omega_c\|^{-2}$ and $\mathbf{x}_0 = e^{T\Delta}\mathbf{x}_1$ to execute perfect high-frequency cancellation, we construct the forward process directly around this physical operator. To prevent exact singularity matching, where the covariance matrix collapses entirely at $t = 0$, we introduce a trace regularizer yielding $\mathbf{x}_0 = e^{T\Delta}\mathbf{x}_1 + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma_{\min}^2 \mathbf{I})$. To completely bypass costly large-kernel spatial convolutions and perfectly execute the mathematically ideal exponential filter, the prior is computed efficiently in the frequency domain. Using the 2D Discrete Cosine Transform (DCT) and the diagonalized Laplacian eigenvalues Λ , the prior is projected as $\mathbf{x}_0 = \text{IDCT}(\exp(T\Lambda) \odot \text{DCT}(\mathbf{x}_1)) + \epsilon$. By strictly implementing the analytical constraints, this construction minimizes the kinetic transport cost E_K while perfectly dissipating the localized Dirichlet energy E_D .

Comparison to IHDMs. Inverse Heat Dissipation Models (IHDMs) [21] attempt to reverse heat dynamics but face fundamentally ill-posed singular trajectories. To prevent the integration from blowing up, they are forced to formulate the reverse process as an SDE, heavily injecting Langevin noise to mask structural singularities. In contrast, our framework operates as a deterministic ODE, eliminating the need for stochastic noise injection. We analytically bound the physical operator using the transient relaxation schedule $\alpha_t = \frac{t}{\|\omega_{\max}\|^2}$. By enforcing $\lim_{t \rightarrow 0} \alpha_t = 0$, the spatial amplification is completely deactivated during the highly ambiguous early stages of generation. As semantic structure solidifies near $t \rightarrow 1$, the schedule α_t smoothly ascends, naturally bypassing the structural singularity without requiring the neural network to act as an SDE denoiser. Consequently, our neural parameterization $\mathbf{v}_\theta$ acts as a pure semantic optimal transport router. This decoupling preserves strict ℓ^2 -Wasserstein transport paths and reduces inference network evaluations by orders of magnitude compared to the expensive SDEs required by IHDMs.

From Vector Field Regression to Clean Data Prediction. Recent theoretical analyses demonstrate that standard continuous-time models predicting off-manifold vector targets $\mathbf{v}$ suffer from high-dimensional capacity bottlenecks, whereas directly predicting clean data ($\mathbf{x}$ -prediction) forces networks to respect the low-dimensional data manifold [13]. However, raw $\mathbf{x}$ -prediction in pixel space typically fails at high resolutions due to the intractable burden of resolving sharp, high-frequency boundary statistics. To transition from $\mathbf{v}$ -prediction to $\mathbf{x}$ -prediction, we define a smoothed manifold as $\tilde{\mathbf{x}}_1 = \mathbf{x}_1 + (1 - t)\alpha_t \Delta \mathbf{x}_t$ and its corresponding ODE

$$\frac{d\mathbf{x}_t}{dt} = \frac{\tilde{\mathbf{x}}_1 - \mathbf{x}_t}{1 - t} - \alpha_t \Delta \mathbf{x}_t. \quad (5)$$

By substituting the $\mathbf{x}$ -prediction parameterization into the original objective, the training loss becomes

$$\mathcal{L}_{\text{DEFM}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[\lambda(t) \|\mathbf{x}_\theta(\mathbf{x}_t, t) - \tilde{\mathbf{x}}_1\|_2^2 \right], \quad (6)$$

where $\lambda(t) = 1/(1 - t)^2$ is the temporal weighting factor. Our Dual-Energy framework structurally resolves this limitation by shifting the neural target to a physically smoothed manifold, formally proven in Appendix C.

Theorem 3. *In the kinematic $\mathbf{x}$ -prediction parameterization of Dual-Energy Flow Matching, the effective neural target is shifted from the raw data $\mathbf{x}_1$ to a smoothed manifold $\tilde{\mathbf{x}}_1 = \mathbf{x}_1 + (1-t)\alpha_t\Delta\mathbf{x}_t$. In the spatial frequency domain, this analytically dampens the high-frequency spectra of the empirical data by a factor of exactly $(1 - t^2(1-t)\|\omega\|^2/\|\omega_{\max}\|^2)$. This guarantees that the neural network regresses onto a strictly band-limited topological manifold, thereby eliminating the need to learn geometric boundaries from scratch.*

In standard flow matching, $\mathbf{x}$ -prediction forces the neural network to regress the exact, sharp empirical data $\mathbf{x}_1$. In contrast, Theorem 3 proves that our objective shifts the neural target to a physically smoothed data manifold. The neural network acts as a low-frequency semantic router, mathematically relieved of the intractable task of generating geometric edges from scratch. During inference, the unified ODE seamlessly reintegrates these physical boundaries via

$$\frac{d\mathbf{x}_t}{dt} = \frac{\mathbf{x}_\theta(\mathbf{x}_t, t) - \mathbf{x}_t}{1-t} - \alpha_t\Delta\mathbf{x}_t. \quad (7)$$

The neural network predicts the smooth topological anchor $\tilde{\mathbf{x}}_1$, while the explicit physical operator $-\alpha_t\Delta\mathbf{x}_t$ acts as a continuous high-pass filter, analytically ascending the Dirichlet energy landscape to perfectly reconstruct the sharp edges in raw pixel space.

5 Conclusion

This work reconceptualizes continuous-time generative modeling from a dual-energy perspective, explicitly separating macroscopic semantic transport from microscopic spatial dynamics. By embedding an analytical discrete spatial Laplacian into the generative vector field, we structurally resolve the core inefficiencies of standard flow matching. First, we formally close the semantic gap, proving that algebraically offsetting high-frequency structural gradients decouples metric entropy from grid resolution, yielding strictly tighter sample complexity bounds. Second, we eliminate the severe numerical stiffness inherent in high-resolution synthesis, mathematically guaranteeing stable deterministic integration without requiring stochastic regularizers. Third, by shifting the $\mathbf{x}$ -prediction target to a physically smoothed data manifold, we relieve the neural network from brute-forcing geometric boundaries, enabling robust, direct pixel-space generation. Ultimately, Dual-Energy Flow Matching establishes a scalable, theoretically grounded framework that explicitly respects the continuous mechanics of the visual domain. Empirical scaling on massive real-world datasets is left as future work, and future research will explore extending these principles into second-order Newtonian dynamics to further stabilize one-step macroscopic synthesis.

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A Proof of Theorem 1

Proof. To track the exact scaling of image resolution and sample complexity on a discrete grid of $M \times M$, we start with the baseline spectral bottleneck, followed by the statistical metric entropy, and conclude with the dual-energy resolution.

The Lipschitz Constraint and Spectral Bottleneck. We establish the required representational capacity of the neural network $\mathbf{v}_\theta$. By definition, a continuously differentiable function is L -Lipschitz if its spatial gradient is universally bounded by L . For an image defined explicitly on a discrete $M \times M$ spatial grid $\mathbf{r} \in \{0, \dots, M-1\}^2$, the neural network $\mathbf{v}_\theta$ must accurately regress the standard optimal transport target $\mathbf{v}^* = \mathbf{x}_1 - \mathbf{x}_0$. To do so without smoothing over sharp geometric edges, the neural network’s hypothesis class $\mathcal{H}_{L_H}$ must possess a Lipschitz constant L_H capable of matching the target’s steepest spatial transitions. Thus, the tightest (minimal) Lipschitz constant L_H required by the neural network is exactly the supremum of the target’s spatial gradient norm

$$L_H = \sup_{\mathbf{r}} \|\nabla_{\mathbf{r}} \mathbf{v}^*(\mathbf{r})\|. \quad (8)$$

To analyze this supremum, we transition to the frequency domain using the 2D Discrete Fourier Transform (DFT), denoted as $\mathcal{F}$. By the finite difference property of the DFT, applying a spatial gradient $\nabla_{\mathbf{r}}$ corresponds to multiplication by the discrete spatial frequency vector ω . The spatial gradient norm is bounded by the discrete spectral summation over all 2D frequencies $\omega \in \mathcal{K}_M$ as

$$L_H = \sup_{\mathbf{r}} \|\nabla_{\mathbf{r}} \mathbf{v}^*(\mathbf{r})\| \leq \sum_{\omega \in \mathcal{K}_M} \|\omega\| \|\mathcal{F}(\mathbf{v}^*)(\omega)\| \quad (9)$$

The discrete frequency domain for an $M \times M$ grid is strictly bounded by the exact Nyquist limit $\|\omega_{\max}\|$. Because the raw empirical data sample $\mathbf{x}_1$ contains sharp edges, it possesses significant signal energy up to this Nyquist limit. The multiplicative factor $\|\omega\|$ in the summation amplifies these high-frequency heavy tails, mathematically driving the minimal required Lipschitz constant to scale linearly with the grid resolution $L_H = \mathcal{O}(M)$.

Metric Entropy and the Curse of Dimensionality. The neural network operates on the flattened image state vector $\mathbf{x} \in \mathbb{R}^{M^2}$. We equip the state space with the maximum norm $\|\cdot\|_\infty$. By the Mean

Value Theorem, bounding the maximum gradient by the minimal constant L_H directly enforces the Lipschitz condition. Any function f in the hypothesis class $\mathcal{H}_{L_H}$ inherently satisfies the condition

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq L_H \|\mathbf{x} - \mathbf{y}\|_\infty. \quad (10)$$

We derive the metric entropy bound for this class. Let the state domain be a compact hypercube $\mathcal{X} = [0, R]^D \subset \mathbb{R}^D$, where the dimensionality is $D = M^2$ and R bounds the pixel intensity range. We bound the number $\mathcal{N}(\epsilon, \mathcal{H}_{L_H}, \|\cdot\|_\infty)$ for a target approximation error $\epsilon > 0$ by establishing both an upper bound and a lower bound.

The Upper Bound. We partition the input domain $\mathcal{X}$ into a grid Σ with spacing $\frac{\epsilon}{2L_H}$, yielding $|\Sigma| = \left(\frac{2RL_H}{\epsilon}\right)^D$ discrete values. We discretize the output range $[-B, B]$ into a grid Γ with spacing $\frac{\epsilon}{2}$, yielding $|\Gamma| = \frac{4B}{\epsilon}$ discrete values. For any $f \in \mathcal{H}_{L_H}$, we construct a discrete approximation $\hat{f} : \Sigma \rightarrow \Gamma$ by rounding $f(\mathbf{x})$ to the nearest value in Γ at each point. Rounding to the nearest value induces a maximum error of $\frac{\epsilon}{4}$. Due to the derived L_∞ Lipschitz constraint, adjacent grid points $\mathbf{x}, \mathbf{y} \in \Sigma$ satisfy $|f(\mathbf{x}) - f(\mathbf{y})| \leq L_H \frac{\epsilon}{2L_H} = \frac{\epsilon}{2}$. Combining this with the rounding error, the discrete function differences are strictly bounded as

$$|\hat{f}(\mathbf{x}) - \hat{f}(\mathbf{y})| \leq |f(\mathbf{x}) - f(\mathbf{y})| + 2\left(\frac{\epsilon}{4}\right) \leq \epsilon. \quad (11)$$

Since the output grid Γ has spacing $\frac{\epsilon}{2}$, this bound restricts the combinatorial branching, meaning knowing $\hat{f}(\mathbf{x})$ leaves at most a constant number of choices C for $\hat{f}(\mathbf{y})$. Constructing a spanning tree through the $|\Sigma|$ points of Σ , the number of valid discretized functions is bounded by $\left(\frac{4B}{C\epsilon}\right)^{|\Sigma|}$. For any arbitrary point $\mathbf{x} \in \mathcal{X}$ with nearest grid point $\mathbf{x}_\Sigma$, the total interpolation error is given by

$$|f(\mathbf{x}) - \hat{f}(\mathbf{x}_\Sigma)| \leq L_H \|\mathbf{x} - \mathbf{x}_\Sigma\|_\infty + \frac{\epsilon}{4} \leq L_H \left(\frac{\epsilon}{4L_H}\right) + \frac{\epsilon}{4} = \frac{\epsilon}{2} < \epsilon. \quad (12)$$

Taking the logarithm gives the metric entropy upper bound $\log \mathcal{N}(\epsilon, \mathcal{H}_{L_H}) \leq \log \left(\frac{4B}{C\epsilon} C^{|\Sigma|}\right) = \mathcal{O}(|\Sigma|) = \mathcal{O}\left(\left(\frac{L_H}{\epsilon}\right)^D\right)$.

The Lower Bound. We partition the domain $\mathcal{X}$ into $K = \left\lfloor \frac{RL_H}{2\epsilon} \right\rfloor^D$ non-overlapping hypercubes of side length $\frac{2\epsilon}{L_H}$. For each sub-cube i with center $\mathbf{c}_i$, we define a local bump function $g_i(\mathbf{x}) = \max(0, \epsilon - L_H \|\mathbf{x} - \mathbf{c}_i\|_\infty)$. Each bump is L_H -Lipschitz, peaks at ϵ , and is entirely confined within its sub-cube. We construct a set by taking linear combinations $f_s(\mathbf{x}) = \sum_{i=1}^K s_i g_i(\mathbf{x})$ with independent binary signs $s_i \in \{0, 1\}$. This yields 2^K distinct globally L_H -Lipschitz functions. Because any two distinct functions f_s and $f_{s'}$ differ in at least one sub-cube j , their maximum divergence evaluates at $\mathbf{c}_j$ as $|f_s(\mathbf{c}_j) - f_{s'}(\mathbf{c}_j)| = |\epsilon - 0| = \epsilon$. The L_∞ distance between any two functions is exactly ϵ . The number must be at least as large as 2^K , yielding $\log \mathcal{N}(\epsilon, \mathcal{H}_{L_H}) \geq K \log 2$. For $\frac{L_H}{\epsilon} \geq \frac{2}{R}$, the floor function scales asymptotically with its argument, establishing the rigorous lower bound $\log \mathcal{N}(\epsilon, \mathcal{H}_{L_H}) = \Omega\left(\left(\frac{L_H}{\epsilon}\right)^D\right)$.

Substituting the spatial dimensionality $D = M^2$, we verify that $\log \mathcal{N}(\epsilon, \mathcal{H}_{L_H}) \propto \left(\frac{L_H}{\epsilon}\right)^{M^2}$. Substituting our resolution-dependent minimal Lipschitz bound $L_H = \mathcal{O}(M)$ into this verified entropy formulation yields the sample complexity bound for standard flow matching $N_{\text{FM}} = \mathcal{O}\left(\left(\frac{M}{\epsilon}\right)^{M^2}\right)$.

The Dual-Energy Resolution and Decoupling. The modified vector field is $\mathbf{v}_\theta = \mathbf{v}^* + \alpha_t \Delta \mathbf{x}_t$. Applying the DFT to $\mathbf{v}_\theta$ yields

$$\mathcal{F}(\mathbf{v}_\theta)(\omega) = \mathcal{F}(\mathbf{v}^*)(\omega) - \alpha_t \|\omega\|^2 \mathcal{F}(\mathbf{x}_t)(\omega). \quad (13)$$

For this algebraic subtraction to effectively cancel the structural gradients, we must guarantee that the prior $\mathbf{x}_0$ introduces zero confounding high-frequency energy. Therefore, the topological prior is strictly derived by dissipating the empirical data through the forward heat semigroup up to a temporal horizon T , giving $\mathbf{x}_0 = e^{T\Delta} \mathbf{x}_1$. Because the eigenfunctions of the discrete Laplacian correspond to the DFT basis elements with eigenvalues $\lambda(\omega) = -\|\omega\|^2$, this operation applies a strict exponential low-pass filter in the frequency domain $\mathcal{F}(\mathbf{x}_0)(\omega) = e^{-T\|\omega\|^2} \mathcal{F}(\mathbf{x}_1)(\omega)$. By defining a macroscopic

topological cutoff frequency $\|\omega_c\|$ and setting $T = \|\omega_c\|^{-2}$, energy at and above this structural cutoff is exponentially suppressed. Because $\mathcal{F}(\mathbf{x}_0) \rightarrow 0$, the high-frequency spectra of both the target vector field and the state are dominated exclusively by the empirical data, yielding $\mathcal{F}(\mathbf{v}^*) \approx \mathcal{F}(\mathbf{x}_1)$ and $\mathcal{F}(\mathbf{x}_t) \approx t\mathcal{F}(\mathbf{x}_1)$. Substituting these asymptotics back into the Fourier-domain neural target isolates the high-frequency residual as

$$\mathcal{F}(\mathbf{v}_\theta)(\omega) \approx \mathcal{F}(\mathbf{x}_1)(\omega) - \alpha_t \|\omega\|^2 (t\mathcal{F}(\mathbf{x}_1)(\omega)), \quad (14)$$

$$\approx \mathcal{F}(\mathbf{x}_1)(\omega) (1 - \alpha_t t \|\omega\|^2). \quad (15)$$

To guarantee that this algebraic term acts as a strict structural dampener on the $M \times M$ grid, we fix the physical relaxation schedule α_t relative to the exact Nyquist limit $\|\omega_{\max}\|$. By setting $\alpha_t = \frac{t}{\|\omega_{\max}\|^2}$, the high-frequency multiplier evaluates exactly to $\left(1 - t^2 \frac{\|\omega\|^2}{\|\omega_{\max}\|^2}\right)$. By substituting this dampened target into the initial Lipschitz summation, the linear derivative penalty $\|\omega\|$ is neutralized at the upper frequency limits as

$$L_L \leq \sum_{\omega \in \mathcal{K}_M} \|\omega\| \left\| \mathcal{F}(\mathbf{x}_1)(\omega) \left(1 - t^2 \frac{\|\omega\|^2}{\|\omega_{\max}\|^2}\right) \right\|. \quad (16)$$

Because the heaviest spectral tails are suppressed by the analytical Laplacian, the new minimal Lipschitz constant L_L is strictly less than L_H and is entirely decoupled from the grid resolution M . Substituting this decoupled L_L back into the metric entropy formulation exponentially shrinks the required hypothesis space, collapsing the intractable sample complexity to a strictly tighter, highly scalable bound given by $N_{\text{Dual}} = \mathcal{O}\left(\left(\frac{L_L}{\epsilon}\right)^{M^2}\right)$. $\square$

B Proof of Theorem 2

Proof. Consider an explicit numerical solver such as Forward Euler integrating a generic generative ODE $\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t)$. Absolute numerical stability requires the step size Δt to satisfy $\Delta t \leq \frac{2}{\rho(\mathbf{J})}$, where $\rho(\mathbf{J})$ is the spectral radius, representing the maximum absolute eigenvalue magnitude of the system’s Jacobian matrix $\mathbf{J} = \nabla_{\mathbf{x}} \mathbf{f}$.

In standard flow matching, synthesizing sharp spatial boundaries requires the unconstrained neural vector field $\mathbf{v}_\theta$ to implicitly learn reverse heat flow mechanics. To drive structural sharpening, its internal Jacobian must approximate the negative spatial Laplacian $\mathbf{J}_{\text{FM}} = \nabla_{\mathbf{x}} \mathbf{v}_\theta \approx -\Delta$. On a discrete $M \times M$ grid, the discrete Laplacian’s spectral radius scales heavily with the spatial resolution as $\rho(\Delta) = \|\omega_{\max}\|^2 \propto M^2$. Consequently, the spectral radius of the neural network’s Jacobian explodes, $\rho(\nabla_{\mathbf{x}} \mathbf{v}_\theta) = \mathcal{O}(M^2)$. This renders the standard ODE mathematically stiff, forcing a catastrophic limitation on the integration step size to prevent numerical divergence via $\Delta t \leq \mathcal{O}(M^{-2})$.

In Dual-Energy Flow Matching, the generative vector field is explicitly decomposed into $\mathbf{v}_\theta(\mathbf{x}_t, t) - \alpha_t \Delta \mathbf{x}_t$. The Jacobian of this composite system is linearly separable as $\mathbf{J}_{\text{DE}} = \nabla_{\mathbf{x}} \mathbf{v}_\theta - \alpha_t \Delta$. We analyze the stiff physical operator $-\alpha_t \Delta$. In the discrete spatial frequency domain, the exact eigenvalues of this operator are given by $\lambda_{\text{phys}}(\omega) = \alpha_t \|\omega\|^2$. Substituting our closed-form relaxation schedule $\alpha_t = \frac{t}{\|\omega_{\max}\|^2}$ yields $\lambda_{\text{phys}}(\omega) = t \frac{\|\omega\|^2}{\|\omega_{\max}\|^2}$. Because the generative temporal path is strictly bounded by $t \in [0, 1]$ and any valid discrete spatial frequency is bounded by the exact Nyquist limit $\|\omega\| \leq \|\omega_{\max}\|$, the maximum eigenvalue of the physical operator is mathematically bounded by $\rho(-\alpha_t \Delta) \leq 1$. Furthermore, by the metric entropy guarantees established in Theorem 1, the explicit Laplacian subtraction guarantees that the neural router $\mathbf{v}_\theta$ is entirely band-limited and smooth. This yields a bounded, resolution-independent neural spectral radius where $\rho(\nabla_{\mathbf{x}} \mathbf{v}_\theta) \ll \mathcal{O}(M^2)$. Because both components of $\mathbf{J}_{\text{DE}}$ are bounded independently of M , the total spectral radius of the composite vector field is bounded by $\mathcal{O}(1)$. The generative ODE is completely non-stiff, enabling stable integration with standard explicit solvers at massive step sizes $\Delta t = \mathcal{O}(1)$ without requiring exact spectral integrators during inference. $\square$

C Proof of Theorem 3

Proof. To align with the manifold prediction objective, we adopt the kinematic $\mathbf{x}$ -prediction parameterization defined by

$$\mathbf{v}_\theta(\mathbf{x}_t, t) = \frac{\mathbf{x}_\theta(\mathbf{x}_t, t) - \mathbf{x}_t}{1 - t}. \quad (17)$$

Given the linear path $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$, we have $\mathbf{x}_1 - \mathbf{x}_0 = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1 - t}$. The vector field of Dual-Energy is $(\mathbf{x}_1 - \mathbf{x}_0) + \alpha_t \Delta \mathbf{x}_t$. Substituting $\frac{\mathbf{x}_1 - \mathbf{x}_t}{1 - t}$ into Eq. 17 yields

$$\frac{\mathbf{x}_\theta(\mathbf{x}_t, t) - \mathbf{x}_t}{1 - t} = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1 - t} + \alpha_t \Delta \mathbf{x}_t, \quad (18)$$

$$\mathbf{x}_\theta(\mathbf{x}_t, t) = \mathbf{x}_1 + (1 - t)\alpha_t \Delta \mathbf{x}_t. \quad (19)$$

The spatial Laplacian applies a negative quadratic penalty

$$\mathcal{F}(\mathbf{x}_\theta)(\omega) = \mathcal{F}(\mathbf{x}_1)(\omega) - (1 - t)\alpha_t \|\omega\|^2 \mathcal{F}(\mathbf{x}_t)(\omega). \quad (20)$$

From the derivations in Theorem 1, we have $\mathcal{F}(\mathbf{x}_t) \approx t\mathcal{F}(\mathbf{x}_1)$. Substituting $\mathcal{F}(\mathbf{x}_t) \approx t\mathcal{F}(\mathbf{x}_1)$ and $\alpha_t = \frac{t}{\|\omega_{\max}\|^2}$ into Eq. 20 yields

$$\mathcal{F}(\mathbf{x}_\theta)(\omega) \approx \mathcal{F}(\mathbf{x}_1)(\omega) - (1 - t) \left(\frac{t}{\|\omega_{\max}\|^2} \right) \|\omega\|^2 (t\mathcal{F}(\mathbf{x}_1)(\omega)), \quad (21)$$

$$\approx \left(1 - t^2(1 - t) \frac{\|\omega\|^2}{\|\omega_{\max}\|^2} \right) \mathcal{F}(\mathbf{x}_1)(\omega). \quad (22)$$

Because $t^2(1 - t) < 1$ and $\|\omega\| \leq \|\omega_{\max}\|$, $\mathbf{x}_\theta$ is guaranteed to contain strictly less high-frequency energy than the raw data $\mathbf{x}_1$. $\square$